CHAPTER 2

Some Basic Concepts of Chemistry

Sr. No.	Concept	Formulas	Other information
1.	**Atomic weight**	Thus, atomic weight = $\boxed{\dfrac{\text{Weight of an atom of the element}}{\text{Weight of an atom of C(Mass no. 12)}}\times 12}$	Here, C is the carbon element
2.	**Average atomic mass**	$\boxed{\begin{array}{l}\text{the average atomic mass of}\\ X=\dfrac{a_1x_1+a_2x_2+.....+a_nx_n}{100}\end{array}}$	Here, x_1, x_2 and x_n relative abundance (%) of first, second and n^{th} isotopes of the element a_1, a_2 and a_n are the atomic masses of isotopes of element respectively
3.	**Number of moles**	Number of moles of atoms $=\dfrac{\text{Number of atoms}}{N_A}$ Number of moles of molecules $=\dfrac{\text{Number of molecules}}{N_A}$ $n=\dfrac{\text{volume at S.T.P. (in litre)}}{22.4\text{ litre}}$ or $\dfrac{\text{Volume at S.T.P. (in mL)}}{22400\text{ mL}}$	Here, n is number of moles, N_A is Avogadro constant $N_A = 6.022 \times 10^{23}$ atoms / molecules / ions
4.	**Weight fraction**	$\text{Weight Fraction}=\dfrac{\text{weight of solute}}{\text{weight of solution}}$	
5.	**Mass percentage (w/w)**	$\text{Mass \% of solute}=\dfrac{\text{Wt. of the solute (in g)}}{\text{Wt. of the solution (in g)}}\times 100$	
6.	**Volume percentage (V/V)**	$\text{Volume \% of solute}=\dfrac{\text{Vol. of solute (in cc)}}{\text{Vol. of solution (in cc)}}\times 100$	

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7.	**Mass by volume percentage (w/V)**	$\text{w/V \% of solute} = \dfrac{\text{Wt. of solute (in g)}}{\text{Vol. of solution (in cc)}} \times 100$	
8.	**Volume fraction**	$\text{Volume Fraction} = \dfrac{\text{Volume of solute}}{\text{Volume of solution}}$	
9.	**Molarity (M)**	$\text{Molarity (M)} = \dfrac{\text{No. of moles of solute}}{\text{Volume of solution in liters}}$ $\text{Molarity} = \dfrac{\text{Moles of the solute}}{\text{Vol. of the solution (in cc)}} \times 1000$ $\text{Molarity} = \dfrac{10 \times x \times d}{M_1}$	Here, $x = \% \frac{w}{w}$ of solute d = density of solution in g/mL M_1 = molar mass of solute
10.	**Molarity for dilute solution**	Molarity for dilute solutions: $M_1V_1 = M_2V_2$	Here, M_1 is molarity before dilution, M_2 is molarity after dilution, V_1 & V_2 is volume before and after dilution
11.	**For mixed solution molarity**	For mixed solution molarity, $= \dfrac{M_1V_1 + M_2V_2 + ...}{V_1 + V_2 + ...}$	Here M_1 is molarity of first solution, M_2 is molarity of second solution, V_1 & V_2 is volume of first and second solution
12.	**Relation between molarity (M) and molality (m)**	$m = \dfrac{1000M}{1000d - MM_1}$	Here, m is molality of the solution M is molarity of the solution M_1 is molar mass of solute d is density of the solution (mg/mL)
13.	**Relation between molarity (M) and mole fraction (χ)**	$M = \dfrac{\chi_B \times 1000 \times d}{\chi_A m_A + \chi_B m_B}$	Here, m_A and m_B are molar masses of solvent and solute respectively. d is density of solution in mg/mL M is molarity of the solution χ_A is mole fraction of solvent χ_B is mole fraction of solute

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14.	**Vapour density**	Vapour density $= \frac{1}{2} \times$ molar mass	
15.	**% Yield of reaction**	% Yield of reaction $= \left\{ \frac{\text{amount of a definite product actually produced in a reaction}}{\text{maximum possible amount of the same product which can be produced}} \right\} \times 100$	
16.	**Mole fraction (χ)**	Mole fraction of solute in solution, $\chi_2 = \frac{n_2}{n_1 + n_2} = \frac{w_2/M_2}{w_1/M_1 + w_2/M_2}$ Mole fraction of solvent in solution, $\chi_2 = \frac{n_1}{n_1 + n_2} = \frac{w_1/M_1}{w_1/M_1 + w_2/M_2}$	Here, w_1 and M_1 are weight and molar mass of solvent respectively. w_2 and M_2 are weight and molar mass of the solute respectively. n_1 is the mole of solvent n_2 is the mole of solute
17.	**Parts per million (ppm)**	Parts per million (ppm)$= \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6$ or ppm $= \frac{\text{Vol. of solute}}{\text{Vol. of solution}} \times 10^6$	